

Part D: Cyber Threat Research

1. Phishing

Definition:

Phishing is a cyberattack where attackers send fake emails, messages, or websites to trick people into revealing sensitive information such as passwords or bank details.

Real-world Example:

A fake email pretending to be from a bank asks you to click a link and log in to "verify your account."

Prevention Methods:

- Verify the sender before clicking links.
 - Avoid opening suspicious attachments.
 - Enable Multi-Factor Authentication (MFA).
 - Check website URLs carefully.
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2. Malware

Definition:

Malware is harmful software designed to damage a computer, steal data, or disrupt normal operations.

Real-world Example:

Downloading a free application from an untrusted website that secretly installs a virus.

Prevention Methods:

- Install trusted antivirus software.
 - Download software only from official websites.
 - Keep your operating system updated.
 - Avoid suspicious downloads and links.
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3. Ransomware

Definition:

Ransomware is a type of malware that locks or encrypts files and demands payment to restore access.

Real-world Example:

A company's files are encrypted, and attackers demand money to provide the decryption key.

Prevention Methods:

- Back up important files regularly.

- Keep software updated.
 - Do not open suspicious email attachments.
 - Use antivirus and firewall protection.
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4. Social Engineering

Definition:

Social engineering is a technique where attackers manipulate people into revealing confidential information or performing unsafe actions.

Real-world Example:

Someone calls pretending to be an IT technician and asks for your login password.

Prevention Methods:

- Never share passwords with anyone.
 - Verify the identity of callers or emails.
 - Be cautious of urgent or unexpected requests.
 - Attend cybersecurity awareness training.
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5. Data Breach

Definition:

A data breach occurs when unauthorized individuals gain access to sensitive or confidential information.

Real-world Example:

Hackers steal customer names, email addresses, and passwords from a company's database.

Prevention Methods:

- Use strong, unique passwords.
- Enable MFA on accounts.
- Encrypt sensitive data.
- Regularly update software and monitor for suspicious activity.